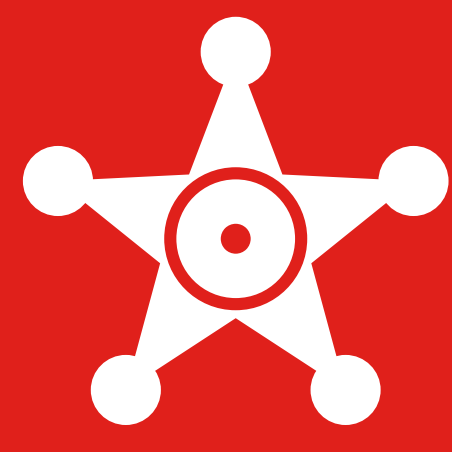
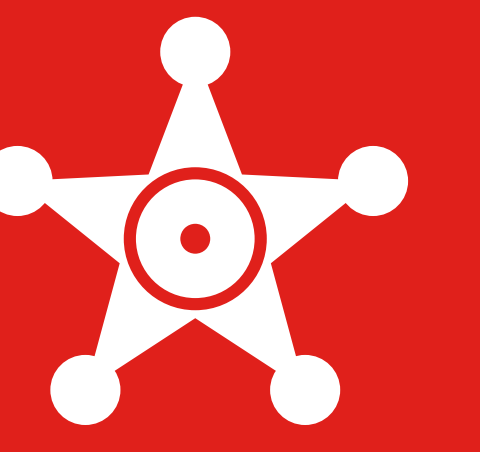




WANTED



ILL-POSED AUTOREGRESSIVE PROBLEMS

THE MODEL

AR(P) with a Gaussian prior on the coefficients is conjugate: the posterior stays closed form.

$$x_t = a_1 x_{t-1} + \dots + a_P x_{t-P} + \varepsilon_t$$

$$\mathbf{a} \sim \text{Normal}(\boldsymbol{\mu}, \mathbf{S})$$

A FILTER IS A POLYNOMIAL

$A(z) = 1 - a_1 z^{-1} - \dots - a_P z^{-P}$ shapes the envelope $|H|^2 = 1/|A|^2$; every expected feature is a factor:

Spectral feature	$Q(z)$	deg
Resonance at (ρ, ω_0)	$1 - 2\rho \cos(\omega_0)z^{-1} + \rho^2 z^{-2}$	2
Real pole at ρ	$1 - \rho z^{-1}$	1
Tilt $-6k$ dB/oct	$(1 - \rho z^{-1})^k$	k
Seasonal pole of period s	$1 - z^{-s}$	s
k -fold resonance	$(1 - 2\rho \cos(\omega_0)z^{-1} + \rho^2 z^{-2})^k$	$2k$

THE UPDATE

The minimum- D_{KL} Gaussian keeps Σ_0 and shifts only the mean onto the constraint:

$$\boldsymbol{\mu}_M = \boldsymbol{\mu}_0 - \Sigma_0 \mathbf{F}^\top (\mathbf{F} \Sigma_0 \mathbf{F}^\top)^{-1} (\mathbf{F} \boldsymbol{\mu}_0 + \mathbf{c})$$

WORKED EXAMPLE

AR(4), impose one resonance $Q(z) \mid A(z)$:

$$A(z) = 1 - a_1 z^{-1} - a_2 z^{-2} - a_3 z^{-3} - a_4 z^{-4}$$

$$Q(z) = 1 - cz^{-1} + dz^{-2}, \quad c = 2\rho \cos \omega_0, d = \rho^2$$

Divisibility is linear in $\mathbf{a}$:

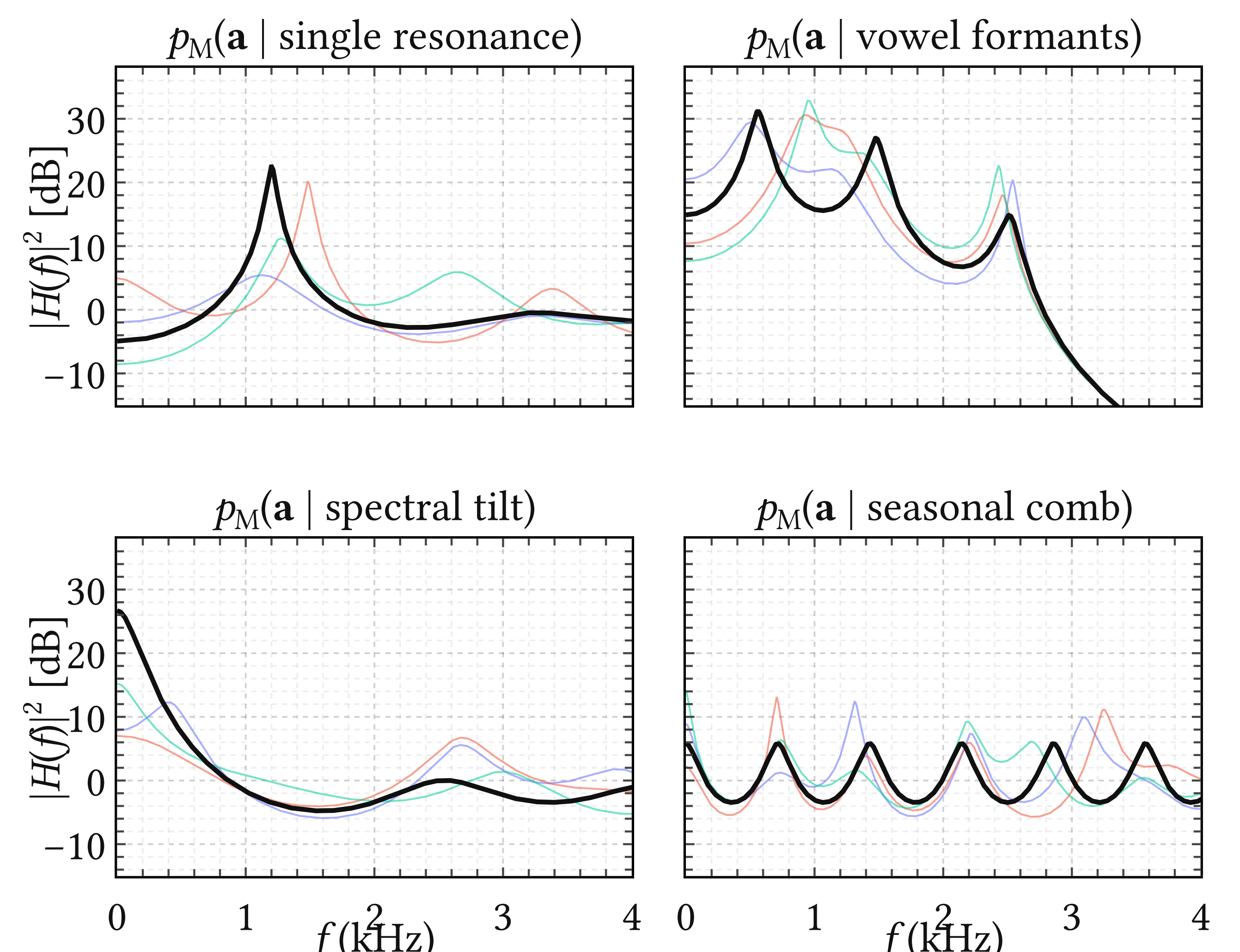
$$Q \mid A \iff \mathbf{F} \mathbf{a} + \mathbf{c} = \mathbf{0}$$

$$\mathbf{F} = \begin{pmatrix} 1 & 0 & -1/d & -c/d^2 \\ 0 & 1 & c/d & c^2/d^2 - 1/d \end{pmatrix}$$

Impose it on *average*, $\mathbb{E}_{\mathbf{a}}[\mathbf{F} \mathbf{a} + \mathbf{c}] = \mathbf{0}$, by the minimum- D_{KL} update of $\mathcal{N}(\mathbf{0}, \lambda \mathbf{I})$: a Gaussian prior with the belief in its mean, uncertainty intact.

BEFORE AND AFTER

The prior before (grey) and after (black) being told about a resonance, three formants, a tilt, a harmonic comb:



THE ASK

You fit AR models: vibration, EEG, radar, seismics, econometrics. You know your resonances, tilts and periods. Tell your filter, and see if it pays.

REWARD

a prior with your knowledge in it, and a co-author who checks the math.

